

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, JULY 2024
Third Semester

21MAB201T – TRANSFORMS AND BOUNDARY VALUE PROBLEMS

(For the candidates admitted from the academic year 2021 – 2022 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (25 × 1 = 25 Marks)

Marks	BL	CO	PO
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Answer **ALL** Questions

- The complete integral of $p=q$ is

(A) $z = ax + by$	(B) $z = a(x + y) + b$
(C) $z = ax + by + c$	(D) $z = ax - by + a$
- The complementary function of $(D^2 + DD' - 2D'^2)z = x^2y$

(A) $z = \phi_1(y - x) + \phi_2(y - 2x)$	(B) $z = \phi_1(y + x) + \phi_2(y + 2x)$
(C) $z = \phi_1(y - x) + \phi_2(y + 2x)$	(D) $z = \phi_1(y + x) + \phi_2(y - 2x)$
- The particular integral of $(D^2 - 2DD')z = e^{2x}$ is

(A) $\frac{e^{-2x}}{2}$	(B) $\frac{xe^{-2x}}{4}$
(C) $\frac{x^2e^{2x}}{2}$	(D) $\frac{x^2e^{2x}}{4}$
- The complete integral of $z = px + qy + 2pq$ is

(A) $z = px + qy + ab$	(B) $z = ax + by + 2ab$
(C) $z = ax + by + 2pq$	(D) $z = px + qy + 2pq$
- The period of the periodic function $\sin x$ is

(A) π	(B) 2π
(C) 3π	(D) 0
- If $f(x)$ is even then $\int_{-a}^a f(x)dx$ is

(A) 0	(B) $2 \int_0^a f(x)dx$
(C) $\int_{-a}^0 f(x)dx$	(D) $\int_0^a f(x)dx$

7. The RMS value of $f(x)=x$ in $-1 \leq x \leq 1$ is 1 1 2 1,2
 (A) 1 (B) 0
 (C) $\frac{1}{\sqrt{3}}$ (D) -1
8. Half range cosine series for $f(x)$ in $(0, \pi)$ is 1 1 2 1,2
 (A) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$ (B) $\frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$
 (C) $\sum_{n=1}^{\infty} b_n \sin nx$ (D) $\sum_{n=1}^{\infty} a_n \cos nx$
9. In wave equation $\frac{\partial^2 y}{\partial t^2} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$ where α^2 stands for 1 1 3 1,2
 (A) $\frac{T}{m}$ (B) $\frac{k}{c}$
 (C) $\frac{m}{T}$ (D) $\frac{k}{m}$
10. One dimensional heat equation is used to find 1 1 3 1,2
 (A) Density (B) Temperature distribution
 (C) Time (D) Displacement
11. A string is stretched between two fixed points $x=0$ and $x=l$. The initial conditions are 1 1 3 1,2
 (A) $y(0, t) = y(x, t) = 0$ (B) $y(x, 0) = \frac{\partial y}{\partial t}(x, 0) = 0$
 (C) $y(0, t) = y(l, t) = 0$ (D) $\frac{\partial y}{\partial t}(0, t) = \frac{\partial y}{\partial t}(l, t) = 0$
12. How many initial and boundary conditions are required to solve 1 1 3 1,2
 $\frac{\partial^2 y}{\partial t^2} = \alpha^2 \frac{\partial^2 y}{\partial x^2}$
 (A) Two (B) Three
 (C) Five (D) Four
13. The Fourier transform of a function $f(x)$ is 1 1 4 1,2
 (A) $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ist} dt$ (B) $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx$
 (C) $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{isx} dx$ (D) $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(s) e^{isx} dx$

14. $F[f(x)\cos ax] =$ 1 1 4 1,2
 (A) $\frac{1}{2}[F(a) + F(s-a)]$ (B) $\frac{1}{2}[F(as) + F(s+a)]$
 (C) $\frac{1}{2}[F(s+a) + F(s-a)]$ (D) $\frac{1}{2}[F(s+a) - F(s-a)]$
15. $F[f(x)*g(x)] =$ 1 1 4 1,2
 (A) $F(s) + G(s)$ (B) $F(s) - G(s)$
 (C) $F(s)G(s)$ (D) $F(s)/G(s)$
16. Under Fourier cosine transform $f(x) = \frac{1}{\sqrt{x}}$ is 1 1 4 1,2
 (A) Complex (B) Inverse function
 (C) Cosine function (D) Self-reciprocal function
17. Radius of curvature of $z[a^n]$ is 1 1 5 1,2
 (A) $|z| < a$ (B) $|z| > a$
 (C) $|z| > |a|$ (D) $|z| < |a|$
18. $z[e^{-5n}] =$ 1 1 5 1,2
 (A) $\frac{z}{z + e^{-5}}$ (B) $\frac{z}{z - e^{-5}}$
 (C) $\frac{z}{z - e^{-1}}$ (D) $\frac{z}{z + e^{-1}}$
19. $Z^{-1}\left[\frac{1}{z-a}\right]$ 1 1 5 1,2
 (A) na^{n-1} (B) a^{n+1}
 (C) a^{n-1} (D) na^n
20. The difference equation formed by eliminating 'a' in $u_n = a2^{n+1}$ is 1 1 5 1,2
 (A) $u_{n+1} - 2u_n = 0$ (B) $u_{n+1} = 0$
 (C) $u_{n+1} - u_n = 0$ (D) $u_n = 0$

PART - B (5 × 8 = 40 Marks)

Answer ALL Questions

21. a. Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$. 8 2 1 1,2
- (OR)
- b. Solve $\left(D^2 + 3DD' - 4D'^2\right)z = \sin y$. 8 2 1 1,2

22. a. Express $f(x) = (\pi - x)^2$ as a Fourier series of period 2π in the interval $0 < x < 2\pi$. 8 3 2 1,2

(OR)

- b. Expand $f(x) = x$ as a half range cosine series in $(0, l)$. 8 3 2 1,2
23. a. A tightly stretched string of length l has its ends fastened at $x=0, x=l$. At $t=0$, the string is in the form $f(x) = k(lx - x^2)$ and then released. Find the displacement at any point on the string at a distance x from one end and at any time $t > 0$. 8 3 3 1,2

(OR)

- b. Heat flows through a uniform bar of length l which has its side insulated and the temperature at the ends kept at zero. If the initial temperature at the interior points of the bar is given by $x^3, 0 < x < l$. Find the temperature distribution in the bar at time t . 8 3 3 1,2
24. a. Evaluate $\int_0^\infty \frac{dx}{(x^2 + a^2)(x^2 + b^2)}$ using Fourier transform methods. 8 3 4 1,2

(OR)

- b. Use Parseval's identity and evaluate $\int_0^\infty \frac{x^2 dx}{(x^2 + a^2)^2}$. 8 3 4 1,2
25. a. Find the Z-transform of $\frac{1}{(n+1)(n+2)}, n > 0$. 8 2 5 1,2

(OR)

- b. Find $Z^{-1} \left[\frac{z^2 + z}{(z-1)^3} \right]$ by long division method. 8 2 5 1,2

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Questions

26. Determine the first three harmonics of the Fourier series for the following data. 15 4 2 1,2

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$
y	1.98	1.30	1.05	1.30	-0.88	-0.25

27. A rod 30 cm long has its ends A and B kept at 20°C and 80°C respectively, until steady state conditions prevail. The temperature at each end is then suddenly reduced to 0°C and kept so. Find the resulting temperature function $u(x, t)$, taking $x=0$ at A. 15 4 3 1,2

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